# HOTEL BOOKING DEMAND

*Executive Dashboard Portfolio*

Dynamic Pricing & Booking Optimization · Analytics Team Lead · July 2026

Repo: github.com/KANguyen2802/Hotel-Booking-Demand

Stack (dashboard): Python, pandas, Streamlit, Plotly, HTML/JS, Power BI (in progress)

## 1. Executive Summary

This portfolio piece documents the executive dashboard layer of the Hotel Booking Demand project: decision-support views that translate 82,811 City and Resort bookings into RevPAR, cancellation, and pricing what-if intelligence for GM / Revenue / Finance stakeholders.

The dashboards sit on a star-schema CSV layer and are delivered in three channels: Streamlit local web (delivered), HTML/JS local web (delivered), and Power BI executive pack (in progress).

Analytics behind the screens produced an asymmetric pricing playbook (City Peak can harden ADR; Resort Peak must not take pure +ADR shock), a cancellation-tier booking workflow, and modeled annualized uplift scenarios of roughly €10k / €59k / €70–85k on a ~€2.84M portfolio proxy—reported as floor / base / upside, not live P&L.

## 2. Project Scope & Objectives

### Scope

- Build interactive executive dashboards for two properties (City Hotel, Resort Hotel) over Jul 2015–Aug 2017.
- Expose portfolio KPIs (bookings, revenue, ADR, occupancy, RevPAR, cancel rate) with hotel/year filters.
- Support deeper exploration: seasonality, segment/country mix, cancellation drivers, and pricing what-if simulation.
- Align visual narrative with the recommend-only pricing playbook and pilot governance (shadow → pilot → scale).

### Objectives

1. Decision clarity: Make City vs Resort demand asymmetry visible in one executive surface.
2. Operational readiness: Give RM / FO / Finance a shared KPI language for the 16-week pilot.
3. Metric integrity: Compute RevPAR / ADR / occupancy with an explicit, auditable formula.
4. Multi-channel delivery: Local web for analysts + static HTML for lightweight sharing + Power BI for stakeholder publish (in progress).

### Out of scope

- Live OTA price push / RMS automation.
- Causal A/B in production (dashboards support shadow/pilot monitoring design only).

## 3. Business Questions Addressed

| # | Business question | Dashboard response |
|---|---|---|
| Q1 | How do City and Resort differ on RevPAR, ADR, occupancy? | Overview + RevPAR with hotel/year filters |
| Q2 | Where does cancellation risk concentrate? | Cancellation Analysis drivers |
| Q3 | What if we move ADR in a controlled band? | Pricing Simulator what-if |
| Q4 | Which markets/segments drive revenue vs cancel? | Segment & country mix |
| Q5 | Ready to pilot asymmetric BAR rules? | KPI pack: ΔRevPAR, Δcancel, walk, Direct |

## 4. Design Thinking Approach

The dashboard workstream combined classic Design Thinking with analytics and BI frameworks that match how this project ships decisions: property-asymmetric pricing, cancel-tier booking, and recommend-only pilots.

### Framework stack (why each one)

| Framework | Role in this project |
|---|---|
| Design Thinking | Outer loop: stakeholder pain → problem framing → dashboard prototypes |
| Double Diamond | Separates problem space (City ≠ Resort) from solution space (playbook + dashboards) |
| CRISP-DM | Structures analytics pipeline feeding the UI (Understanding → Modeling → Evaluation → recommend-only deploy) |
| Jobs-to-be-Done | Defines what GM / RM / FO / Finance hire the dashboard to do |
| Decision-centric IA | Page flow: Overview → Diagnose → Act (Simulator) |
| Shneiderman mantra | Overview first, zoom & filter, then details-on-demand |

### Jobs-to-be-Done (stakeholder jobs)

| Stakeholder | Job the dashboard must help them do |
|---|---|
| GM | See portfolio health and City vs Resort trade-offs in one view |
| Revenue Management | Inspect RevPAR / ADR / seasonality and stress-test BAR moves in band |
| Front Office / Ops | See where cancel risk concentrates before buffer / walk decisions |
| Finance | Trace KPI definitions; treat uplift as proxy, not committed P&L |

### Double Diamond × Design Thinking

1. Discover / Empathize — Decision pain: raise RevPAR without uncontrolled cancel, walk, or OTA conflict when City and Resort do not share the same demand curve.
2. Define — Lock problem to property-asymmetric demand + cancel-tier inventory; reject one portfolio elasticity and reject live push of ~+21% City RAISE.
3. Develop / Ideate + Prototype — Map CRISP-DM outputs (scores, ε, ensemble bands) to four decision pages and three channels on the same star-schema facts.
4. Deliver / Test — Validate filters, weighted KPIs, and recommend-only policy (what-if inside ±15% band; no auto-publish).

### CRISP-DM → dashboard handoff

| CRISP-DM phase | Project artifact | Dashboard exposes |
|---|---|---|
| Business Understanding | Exec RevPAR vs risk question | Overview narrative KPIs |
| Data Prep | v5 → star-schema CSV | Stable facts for filters |
| Modeling | LightGBM, ε, BAR ensemble | Cancellation + Pricing inputs |
| Evaluation | Back-test go/no-go, dual-objective | Guardrails in what-if |
| Deployment | Playbook + 16-week pilot | ΔRevPAR, Δcancel, walk, Direct language |

### Decision-centric information architecture

| Layer | Pages | Intent |
|---|---|---|
| Overview | Home | Portfolio pulse — bookings, revenue, ADR, Occ, RevPAR, cancel |
| Diagnose | RevPAR · Cancellation | Seasonality, mix, cancel drivers |
| Act (recommend-only) | Pricing Simulator | Controlled ADR what-if — not RMS write-back |

North-star design principle: show the trade-off (ADR vs volume vs cancel risk), not a single “optimal price” number—consistent with ensemble floor–recommend–ceil and dual-objective softening of pure-revenue optima.

## 5. Dataset Overview

| Attribute | Detail |
|---|---|
| Source file | hotel_bookings_v5.csv (cleaned panel) |
| Grain | 1 row = 1 booking |
| Volume | 82,811 bookings |

| | |
|---|---|
| Properties | City Hotel, Resort Hotel |
| Time window | Arrival months 2015-07 → 2017-08 |
| Overall cancel rate | 28.12% |
| High-risk cancel tier | P ≥ 0.55 → realized cancel ~64% (n = 28,560) |

## 6. Data Model Design

### Schema Choice

Star schema (CSV exports) chosen over a single denormalized flat file for dashboards.

| Layer | Files | Grain |
|---|---|---|
| Fact (monthly) | revpar_monthly.csv | Hotel × calendar month |
| Fact (booking) | hotel_bookings_normalized.csv | 1 booking |
| Dimensions | dim_hotel, dim_date, dim_country, … | Lookups for Power BI / joins |

Why star schema: clear separation of monthly RevPAR KPIs vs booking-level cancel/segment analysis; stable grain for Power BI; rebuildable via scripts/build_star_schema_v5.py (pandas).

## 7. Dashboard Architecture

hotel_bookings_v5.csv → build_star_schema_v5.py → star-schema CSVs → Streamlit (dashboard/) | HTML JSON export (dashboard-html/) | Power BI (dashboard-powerbi/, in progress).

### Streamlit pages

| Page | Purpose |
|---|---|
| Overview | CEO KPIs, revenue/bookings, status, segment, country |
| RevPAR | ADR × occupancy, seasonality, monthly panel |
| Cancellation Analysis | Deposit / channel / segment / lead-time drivers |
| Pricing Simulator | Monthly what-if ADR moves (recommend-only) |

Shared UX: sidebar Hotel/Year multiselect; weighted portfolio KPIs; design tokens from design-system/hotel-booking-demand/.

## 8. Key Findings & Strategic Recommendations

### Key findings

1. City Peak tolerates controlled ADR increase: +10% ADR → +2.3% RevPAR; Peak back-test win-rate 100% in the evaluated window.
2. Resort Peak does not: same +10% ADR → −2.1% RevPAR — pure ADR shock is policy-forbidden.
3. Pure analytic optima (~+21% City RAISE) are too aggressive; operable ±15% bands plus dual-objective softens BAR by ~7–8%.
4. Cancellation is manageable by tier: High-risk bookings feed buffer → Direct @ BAR instead of OTA dump.
5. Narrow rules back-tested 2015–2016 with go=True (ΔRevPAR ≥ 0; Δcancel ≤ +1 pp) for both properties.

### Strategic recommendations

| ID | Action |
|---|---|
| R1 | Asymmetric playbook: harden City Peak; HOLD Resort Peak; Resort Low CUT ~−5% |
| R2 | Never publish extreme +21% as live BAR; keep ensemble band + risk controls |
| R3 | Route bookings by cancel tier (Low / Med / High→Direct) |
| R4 | Prefer Direct refill at recommend BAR before OTA dump |
| R5 | 16-week staged pilot with kill switches (walk >5%/week; cancel >+1 pp) |

Modeled impact (proxy): Conservative ~€10k · Full in-band ~€59k · Direct/buffer upside ~€70–85k on ~€2.84M base.

## 9. Technical Implementation Notes

| Topic | Implementation |
| --- | --- |
| Runtime data access | dashboard/lib/db.py — pandas reads star-schema CSVs (cached) |
| Charts | Plotly (Streamlit); Chart.js / custom JS (HTML) |
| HTML data refresh | dashboard-html/_export_data.py → JSON aggregates |
| Local HTML serve | python -m http.server (no file:// fetch) |
| Power BI | dashboard-powerbi/ — PBIX in progress |
| Dependencies | streamlit, plotly, pandas |
| Rebuild facts | python scripts/build_star_schema_v5.py |

## 10. Metric Calculation Integrity Notes

| Metric | Formula / rule |
| --- | --- |
| Occupancy (hotel-month) | mean(1 − is_canceled) |
| ADR (hotel-month) | mean(adr) where is_canceled = 0 |
| RevPAR (hotel-month) | ADR × Occupancy_Rate |
| Cancel rate (filter) | canceled_bookings / total_bookings |
| Overview ADR/Occ/RevPAR | Booking-weighted across months in active filter |

### Integrity controls

- Monthly RevPAR is recomputed from booking facts, not an opaque external field.
- City and Resort are never blended into one elasticity on decision pages.
- Pricing Simulator is what-if / recommend-only; it does not write prices to channels.
- ROI figures in reports are counterfactual proxies—not committed P&L.

### Known limitations

- Panel ends 2017-08 — recalibrate before live scale.
- No production RCT yet — shadow ≥ 2 weeks required before price pilot.
- Power BI executive pack is in progress; Streamlit and HTML are current interactive deliverables.

*Portfolio document for the dashboard workstream of Hotel Booking Demand · July 2026*